6.4 Projective Varieties

6.4.1 Projective Algebraic Sets

Proposition 6.4.1 (Projective Space is Functorial)
Let $\phi : K \to L$ be an injective ring homomorphism Then there is a well-defined injective map

$$\begin{aligned} \mathbb{P}^n(K) &\to \mathbb{P}^n(L) \\ (x_0 : ... : x_n) &\to (\phi(x_0) : ... : \phi(x_n)) \end{aligned}$$

Proof. Evidently the map is well-defined. Suppose that

$$(\phi(x_0) : ... : \phi(x_n)) \sim (\phi(x_0') : ... : \phi(x_n'))$$

Then for some i we have $x_i \neq 0$ and $\phi(x_i) = \lambda\phi(x_i')$ and $\lambda \in L^\star$. Therefore $\phi(x_i') \neq 0 \implies x_i' \neq 0$ and $\lambda = \phi(x_i/x') =: \phi(\mu)$. Similarly $\phi(x_j) = \phi(\mu)\phi(x_j') \implies \phi(x_j - \mu x_j') = 0 \implies x_j = \mu x_j'$. ∎

Proposition 6.4.2 (Zero Loci in Projective Space)
*Let $A = k[X_0, ..., X_n]$ be the graded polynomial ring in $(n+1)$-variables over a field k. For a homogenous ideal $\mathfrak{a} \subset k[X_0, ..., X_n]$ and field extension K/k define the **zero-locus***

$$V_+(\mathfrak{a})(K) \coloneqq \{(x_0 : ... : x_n) \in \mathbb{P}^n(K) \mid F(x_0, ..., x_n) = 0 \quad \forall F \in \mathfrak{a}\}$$

Similarly for a subset $Y \subset \mathbb{P}^n(K)$ define

$$I_+(Y) \coloneqq \{F \in k[X_0, ..., X_n] \mid F_d(x_0, ..., x_n) = 0 \quad \forall d \geq 0, (x_0 : ... : x_n) \in Y\}$$

Then

a) $I_+(Y)$ is a homogenous radical ideal and $\sqrt{\langle S \rangle} \subseteq I_+(V_+(S)(K))$
b) I_+ and V_+ are order-reversing
c) $S \subseteq I_+(V_+(S)(K))$ and $Y \subseteq V_+(I_+(Y))(K)$
d) $V_+ \circ I_+ \circ V_+ = V_+$ and $I_+ \circ V_+ \circ I_+ = I_+$
e) $V_+(S) = V_+\left(\sqrt{\langle S \rangle}\right)$ and $V_+(\mathfrak{a}) = V_+(\sqrt{\mathfrak{a}})$
f) $\bigcap_i V_+(S_i) = V_+\left(\bigcup_i S_i\right)$ and $\bigcap_i V_+(\mathfrak{a}_i) = V_+\left(\sum_i \mathfrak{a}_i\right)$
g) $\bigcap_i I_+(W_i) = I_+(\bigcup W_i)$
h) $V_+(\mathfrak{a}) \cup V_+(\mathfrak{b}) = V_+(\mathfrak{a} \cap \mathfrak{b})$
i) $V_+((0)) = \mathbb{P}^n(K)$

*The sets of the form $V_+(\mathfrak{a})(K)$ determine the **Zariski Topology** on $\mathbb{P}^n(K)$. We denote the topological space by $\mathbb{P}_k^n(K)$. The topological closure satisfies the following identities*

$$V_+(I_+(X)) = \overline{X}$$

and

$$I_+(X) = I_+\left(\overline{X}\right)$$

There is a form of the Weak Nullstellensatz

$$V_+(\mathfrak{a})\left(\overline{k}\right) = \emptyset \iff A_+ \not\subset \sqrt{\mathfrak{a}} \iff \mathfrak{a} \text{ irrelevant}$$

and a form of the Strong Nullstellensatz namely

$$\mathfrak{a} \text{ essential} \implies I_+\left(V_+(\mathfrak{a})\left(\overline{k}\right)\right) = \sqrt{\mathfrak{a}}$$

Further there is a dual isomorphism

$$\left\{\mathfrak{a} \lhd A \;\middle|\; \mathfrak{a} \text{ radical homogenous and } A_+ \not\subset \mathfrak{a}\right\} \underset{I_+}{\overset{V_+}{\rightleftarrows}} \left\{Z \in \mathbb{P}_k^n\left(\overline{k}\right) \;\middle|\; Z \neq \emptyset\right\}$$

Proof. Let $V_K(S) \coloneqq \left\{(x) \in K^{n+1} \;\middle|\; f(x) = 0 \quad \forall s \in S\right\}$ denote the affine zero-locus. Then $V_+(S)(K) = \pi(V_K(S) \setminus \{\mathbf{0}\})$, where $\mathbf{0} = (0, ..., 0)$ is the origin in K^{n+1}.

a) Homogeneity follows from definition. The relation is straightforward.
b) Follows by definition
c) The pair I, V form a Galois connection by (2.1.54) and the relations follow formally by (2.1.50).

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d) Follows formally by (2.1.52)
e) Follows from (2.1.53) by considering the closure operators $\sqrt{\langle - \rangle}$ and $\langle - \rangle$.
f) The first equality follows from (2.1.55). The second equality follows from (3.3.29).
g) This follows from (2.1.55).
h) Then using the affine case we may deduce

$$V_+(\mathfrak{a} \cap \mathfrak{b}) = \pi(V(\mathfrak{a}) \cup V(\mathfrak{b}) \setminus \mathbf{0}) = \pi(V(\mathfrak{a}) \setminus \mathbf{0}) \cup \pi(V(\mathfrak{b}) \setminus \mathbf{0}) = V_+(\mathfrak{a}) \cup V_+(\mathfrak{b})$$

i) Trivial

Weak Nullstellensatz. Observe that using the affine case $V_{\overline{k}}(A_+) = \{\mathbf{0}\}$ and $I_{\overline{k}}(\mathbf{0}) = A_+$. Therefore

$$V_+(\mathfrak{a})\left(\overline{k}\right) = \emptyset \iff V_{\overline{k}}(\mathfrak{a}) \subset \{\mathbf{0}\} \overset{(3.12.7)}{\iff} A_+ \not\subset \sqrt{\mathfrak{a}}$$

Strong Nullstellensatz. Assume $\mathfrak{a}$ is an essential homogenous ideal then $Y \coloneqq V(\mathfrak{a}) \setminus \{\mathbf{0}\}$ is non-empty and $\overline{Y} = V(\mathfrak{a})$. Because $\mathfrak{a}$ is homogenous then Y is a cone – $x \in Y \implies \lambda x \in Y$ for $\lambda \in \overline{k}^\star$. We claim that $V_+(\mathfrak{a}) = \pi(Y)$ and $I(Y) = I_+(\pi(Y))$. Suppose $F \in I(Y)$ then $F(\lambda x) = 0$ for all $\lambda \in \overline{k}^\star$ and $x \in Y$. As $\overline{k}$ is infinite we deduce that $F_d \in I(Y)$ whence $F \in I_+(\pi(Y))$. The remaining claims are clear. Therefore we may deduce from (6.1.2) and (6.1.13)

$$I_+V_+(\mathfrak{a}) = I_+(\pi(Y)) = I(Y) = I\left(\overline{Y}\right) = I(V(\mathfrak{a})) = \sqrt{\mathfrak{a}}$$

Topological Closure. By (2.1.52) $V_+ \circ I_+$ is a closure operator with image precisely the closed sets. Therefore by (2.1.41) $V_+ \circ I_+$ is the topological closure (4.1.19). Finally $I_+\left(\overline{X}\right) = I_+(V_+(I_+(X))) = I_+(X)$ using d).

The dual isomorphism follows from the Strong Nullstellensatz. ∎

Definition 6.4.3 (Projective Variety)
For an essential homogenous radical ideal $\mathfrak{a} \lhd k[X_0, ..., X_n]$ we say the family

$$X(K) \coloneqq V_+(\mathfrak{a})(K) \subset \mathbb{P}^n_k(K)$$

*is a **projective variety** and write it as $X(K) \coloneqq V_+(\mathfrak{a})(K)$. We say X is **integral** if $X\left(\overline{k}\right)$ is irreducible in the subspace topology. We also define the **homogenous coordinate ring***

$$k[X]_{\text{hom}} \coloneqq k[X_0, ..., X_n]/\mathfrak{a}$$

*which is naturally a positive graded ring by (3.11.12). Define the **irrelevant ideal***

$$k[X]_+ \coloneqq \bigoplus_{d>0} k[X]_d$$

We may generalise (6.4.2) as follows.

Proposition 6.4.4 (Zero Loci in Projective Varieties)
*Let $X = V_+(\mathfrak{a}) \subset \mathbb{P}^n_k$ be a projective variety and $A = k[X]_{\text{hom}}$ the coordinate ring. For any homogenous ideal $\mathfrak{b} \lhd A$ and field extension K/k define the **zero-locus** by*

$$V_+(\mathfrak{b})(K) \coloneqq \{(x_0 : ... : x_n) \in X(K) \mid f(x_0, ..., x_n) = 0 \quad \forall f \in \mathfrak{b}\}$$

Similarly for a subset $Y \subset X(K)$ define

$$I_+(Y) \coloneqq \{f \in k[X] \mid f_d(x) = 0 \quad \forall d \geq 0, x \in Y\}$$

Suppose $\pi : k[X_0, ..., X_n] \to k[X]$ is the quotient map. Then we have the following properties

$$V_+(\mathfrak{b})(K) = V_+(\pi^{-1}(\mathfrak{b}))(K) \cap X(K)$$
$$I_+(Y \cap X(K)) = \pi(I_+(Y)) \quad Y \subset \mathbb{P}^n_k(K)$$
$$I_+(V_+(\mathfrak{b})(K)) = \pi(I_+(V_+(\pi^{-1}(\mathfrak{b}))))$$

The pair $(V_+(-)(K), I_+)$ satisfies the same properties as (6.4.2). The image of $V_+(-)(K)$ forms the closed sets of a topology on $X(K)$ which coincides with the subspace topology from $\mathbb{P}^n_k(K)$. Further the Weak and Strong Nullstellensatz also hold

$$V_+(\mathfrak{b})\left(\overline{k}\right) = \emptyset \iff k[X]_+ \not\subset \sqrt{\mathfrak{b}} \iff \mathfrak{b} \text{ is irrelevant}$$

and

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$$\mathfrak{b} \text{ essential} \implies I_+\left(V_+(\mathfrak{b})\left(\overline{k}\right)\right) = \sqrt{\mathfrak{b}}$$

Further there is a dual isomorphism

$$\left\{\mathfrak{b} \lhd k[X]_{\text{hom}} \;\middle|\; \mathfrak{b} \text{ radical, homogenous and } A_+ \not\subset \mathfrak{b}\right\} \underset{I_+}{\overset{V_+}{\rightleftarrows}} \left\{Z \in X\left(\overline{k}\right) \;\middle|\; Z \neq \emptyset\right\}$$

Proposition 6.4.5 (Projective Subvariety)
*Let X be a projective variety, $\mathfrak{b} \lhd k[X]_{\text{hom}}$ an essential radical homogenous ideal. Then $V_+(\mathfrak{b})(-)$ is a **closed subvariety** which coincides with the projective variety $V_+(\pi^{-1}(\mathfrak{b}))$. There is a bijective correspondence between non-empty closed subsets of $X\left(\overline{k}\right)$ and closed subvarieties of X.*

Conversely if X', X are two projective varieties such that $X'\left(\overline{k}\right) \subset X\left(\overline{k}\right)$ then X' may be expressed as a closed subvariety.

Proof. There is a canonical graded surjective k-algebra homomorphism $\pi : k[X_0, ..., X_n] \to k[X]_{\text{hom}}$ by (3.11.12). Then $\pi^{-1}(\mathfrak{b})$ is graded (3.11.10) and radical (3.3.50) g). Also by definition of the quotient grading on π we find $\pi^{-1}(k[X]_+) = k[X_0, ..., X_n]_+$ so that $\mathfrak{b}$ essential $\implies \pi^{-1}(\mathfrak{b})$ is essential (3.12.7). The bijective correspondence exists by (6.4.4).

If $X' = V_+(\mathfrak{b}')$, $X = V_+(\mathfrak{b})$ are two projective varieties such that $X'\left(\overline{k}\right) \subset X\left(\overline{k}\right)$ then by the dual isomorphism we have $\mathfrak{b} \subset \mathfrak{b}'$. Therefore X' may be represented as the closed subvariety $V_+(\pi(\mathfrak{b}'))$ ∎

Proposition 6.4.6 (Principal Open Sets)
Let $X \subset \mathbb{P}^n_k$ be a projective variety. The Zariski topology on $X(K)$ has a base consisting of subsets of the form

$$D_+(f)(K) \coloneqq X(K) \setminus V_+((f)) = \{(x) \in X(K) \mid f_d(x) \neq 0 \text{ some } d > 0\}$$

where $f \in k[X]$ is a homogenous form. Note that $(x) = (y) \implies (f_d(x) = 0 \iff f_d(y) = 0)$ so the second form is well-defined.

Proof. A generic open set $U = X(K) \setminus V_+(\mathfrak{b})(K)$ where $\mathfrak{b}$ is a homogenous ideal of $k[X]$. Given $(x) \in U$ there is some $f \in \mathfrak{b}$ and $d > 0$ such that $f_d(x) \neq 0$. Therefore $D_+(f_d) \subset U$ is an open neighbourhood of (x). ∎

Proposition 6.4.7
Let $X = V_+(\mathfrak{b})$ be a projective variety and $f, g \in k[X]$ essential homogenous forms. Then

$$D_+(g) \subset D_+(f) \iff f \mid g^N \text{ some } N > 0$$

Proof.

$$D_+(g) \subset D_+(f) \iff V_+((f)) \subset V_+((g)) \iff \sqrt{(g)} \subset \sqrt{(f)} \iff f \mid g^N$$

∎

Proposition 6.4.8 (Irreducible Subsets)
Let $X = V_+(\mathfrak{a}) \subset \mathbb{P}^n_k$ be a projective variety and $\mathfrak{b} \lhd k[X]$ a homogenous radical ideal. Then a closed subvariety $V_+(\mathfrak{b}) \subset X$ is integral iff $\mathfrak{b}$ is prime.

In particular $\mathbb{P}^n_k\left(\overline{k}\right)$ is integral.

Proof. Let $Y \coloneqq V_+(\mathfrak{b})$. Suppose that Y is not irreducible then $Y \subseteq V_+(\mathfrak{c}) \cup V_+(\mathfrak{d})$ is a non-trivial decomposition into closed subsets for $\mathfrak{c}, \mathfrak{d}$ essential homogenous radical ideals for which $Y \not\subset V_+(\mathfrak{c}) \implies \mathfrak{c} \not\subset \mathfrak{b}$ and similarly $\mathfrak{d} \not\subset \mathfrak{b}$ (see (4.1.61)). Choose $f \in \mathfrak{c} \setminus \mathfrak{b}$ and $g \in \mathfrak{d} \setminus \mathfrak{b}$. Then fg is zero on Y whence $fg \in \mathfrak{b}$. This shows that $\mathfrak{b}$ is not prime.

Conversely suppose $Y \subset V_+(\mathfrak{c}) \cup V_+(\mathfrak{d}) = V_+(\mathfrak{c} \cap \mathfrak{d})$ then by (6.4.2) $\mathfrak{c}\mathfrak{d} \subset \mathfrak{c} \cap \mathfrak{d} \subset \mathfrak{b}$. By (3.3.39) we have $\mathfrak{c} \subset \mathfrak{b}$ or $\mathfrak{d} \subset \mathfrak{b}$, whence $Y \subset V_+(\mathfrak{c})$ or $Y \subset V_+(\mathfrak{d})$. Therefore we conclude that Y is irreducible by (4.1.61).

Therefore $\mathbb{P}^n_k\left(\overline{k}\right)$ is irreducible by the special case both $\mathfrak{a}$ and $\mathfrak{b}$ being the zero ideals. ∎

Proposition 6.4.9 (Dual Isomorphism between Closed Sets and Homogenous Ideals)
Let $X = V_+(\mathfrak{a}) \subset \mathbb{P}^n_k$ be a projective variety and $A = k[X]$ the coordinate ring. Under the dual isomorphism

$$\left\{\mathfrak{b} \lhd A \;\middle|\; \mathfrak{b} \text{ radical homogenous and } A_+ \not\subset \mathfrak{b}\right\} \underset{I_+}{\overset{V_+}{\rightleftarrows}} \left\{Z \subset X\left(\overline{k}\right) \text{ closed} \;\middle|\; Z \neq \emptyset\right\}$$

we have the following correspondence

- *irreducible sets correspond to prime ideals*

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- *irreducible components (of $V_+(\mathfrak{b})$) correspond to minimal prime ideals (of $\mathfrak{b}$)*

Proposition 6.4.10 (Decomposition into Irreducible Components)
Let $X = V_+(\mathfrak{a})$ be a projective variety then the topological space $X\left(\overline{k}\right)$ is Noetherian. Furthermore every closed subset Z has finitely many irreducible components Z_i and the decomposition

$$Z = Z_1 \cup ... \cup Z_n$$

is the unique incomparable decomposition into irreducible closed subsets.

If $Z = V_+(\mathfrak{b})$ then the irreducible components correspond to precisely the minimal prime ideals over $\mathfrak{b}$.

Proof. $X\left(\overline{k}\right)$ is Noetherian by using the order isomorphism in (6.4.9) and the fact $k[X]$ is Noetherian. The result then follows from (4.1.76). ∎

6.4.2 Affine Charts

References:

- Algebraic Curves [Ful08] (Chap. 4.3)
- Algebraic Geometry [Har13] (Chap. I §2)

For every $i = 0, ..., n$ the subset of $\mathbb{P}^n_k(K)$ consisting of points of the form

$$\{(x_0 : ... : x_n) \mid x_i \neq 0\}$$

may be naturally identified with $\mathbb{A}^n_k(K)$. The purpose of this section is to make this notion precise.

Lemma 6.4.11
There is an isomorphism of rings

$$\begin{aligned} \theta_i^\sharp : k[X_0, ..., X_n]_{(X_i)} &\longleftrightarrow k[x_0, ..., x_{i-1}, x_{i+1}, ..., x_n] \\ \frac{X_j}{X_i} &\longleftrightarrow x_j \end{aligned}$$

Proof. The algebra homomorphism $\theta_i^\sharp$ exists by (3.7.6) and (3.5.5), restricting to the subring $k[X_0, ..., X_n]_{(X_i)}$. The inverse algebra homomorphism exists by (3.7.6), sending $x_j \to \frac{X_j}{X_i}$. We may verify directly that these are mutually inverse. ∎

Lemma 6.4.12 ((De-)Homogenisation)
Fix n and $0 \leq i \leq n$. For $G \in k[X_0, ..., X_n]$ homogenous and $F \in k[x_0, ..., x_{i-1}, x_{i+1}, ..., x_n]$ define

$$\begin{aligned} G^a &:= G(x_0, ..., x_{i-1}, 1, x_{i+1}, ..., x_n) \\ F^h &:= X_i^{\deg(F)} F\left(\frac{X_0}{X_i}, ..., \frac{X_{i-1}}{X_i}, \frac{X_{i+1}}{X_i}, ..., \frac{X_n}{X_i}\right) \\ &= \sum_{j=0}^{\deg F} X_i^{\deg(F)-j} F_j(X_0, ..., X_{i-1}, X_{i+1}, ..., X_n) \end{aligned}$$

Then

a) $(FF')^h = F^h F'^h$ and $(GG')^a = G^a G'^a$
b) $F^{ha} = F$ and $X_i^k G^{ah} = G$ where $k := v_{X_i}(G)$ is the smallest power of X_i appearing in G.
c) For $\mathfrak{b} \lhd k[X_0, ..., X_n]$ a homogenous ideal the set $\mathfrak{b}^a := \{G^a \mid G \in \mathfrak{b}\}$ is an ideal.

For $\mathfrak{a} \lhd k[x_0, ..., x_{i-1}, x_{i+1}, ..., x_n]$ define the homogenous ideal

$$\mathfrak{a}^h := \left\langle \{F^h \mid F \in \mathfrak{a}\} \right\rangle$$

Then we also have the identities

d) $\mathfrak{b} \subset \mathfrak{b}^{ah}$ with equality when $\mathfrak{b}$ is radical and $\mathfrak{b} \subset \mathfrak{p}$ minimal $\implies X_i \notin \mathfrak{p}$
e) $\mathfrak{a} = \mathfrak{a}^{ha}$
f) $\mathfrak{b}$ prime and $X_i \notin \mathfrak{b} \implies \mathfrak{b}^a$ prime.
g) $\mathfrak{a}$ prime $\implies \mathfrak{a}^h$ prime
h) $\mathfrak{a}$ radical $\implies \mathfrak{a}^h$ radical

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Proof. Mostly tedious verification

a) Using the notation of (6.4.11) we see that $F^h = X_i^{\deg(F)}\left(\theta_i^\sharp\right)^{-1}(F)$ so evidently $(FF')^h = F^h F'^h$ because $\deg(FF') = \deg(F) + \deg(F')$.

   The second relation follows immediately from (6.4.11).

b) The first identity is clear. We may write

$$G = \sum_{j=0}^{\deg(G)-v_{X_i}(G)} X_i^{\deg(G)-j} G_j(X_0, ..., X_{i-1}, X_{i+1}, ..., X_n)$$

   where $\deg\left(G_j\right) = j$ and $v_{X_i}(G)$ is the smallest power of X_i appearing in G. So

$$G^a = \sum_{j=0}^{\deg(G)-v_{X_i}(G)} G_j(x_0, ..., x_{i-1}, x_{i+1}, ..., x_n)$$

$$G^{ah} = \sum_{j=0}^{\deg(G)-v_{X_i}(G)} X_i^{\deg(G)-v_{X_i}(G)-j} G_j(X_0, ..., X_{i-1}, X_{i+1}, ..., X_n)$$

   whence $G = X_i^{v_{X_i}(G)} G^{ah}$.

c) This is evident because $(-)^a$ is a ring homomorphism.

d) The first part is evident from b). Suppose $G \in \mathfrak{b}^{ah}$ then

$$G = \sum_{j=1}^{r} \lambda_j G_j^{ah} \Longrightarrow X_i^k G = \sum_{j=1}^{r} \lambda_j X_i^{k_j} G_j$$

   for $G_j \in \mathfrak{b}$. Then $X_i^k G \in \mathfrak{b} \Longrightarrow G \in \mathfrak{b}$ as we assumed $X_i \notin \mathfrak{b}$.

e) Trivially $F \in \mathfrak{a} \Longrightarrow F^h \in \mathfrak{a}^h \Longrightarrow F^{ha} = F \in \mathfrak{a}^{ha}$. Conversely any $G \in \mathfrak{a}^h$ is of the form

$$G = \sum_j \lambda_j F_j^h \Longrightarrow G^a = \sum_j \lambda_j^a F_j^{ha} = \sum_j \lambda_j^a F_j \in \mathfrak{a}^{ha}$$

   for some $\lambda_i \in k[X_0, ..., X_n]$.

f) Suppose $FF' \in \mathfrak{b}^a \Longrightarrow F^h F'^h \in \mathfrak{b}^{ah} = \mathfrak{b} \Longrightarrow F^h \in \mathfrak{b}$ or $F'^h \in \mathfrak{b} \Longrightarrow F \in \mathfrak{b}^a$ or $F' \in \mathfrak{b}^a$.

g) Suppose $GG' \in \mathfrak{a}^h$ then $GG' = \sum_j \lambda_j F_j^h$ for $F_j \in \mathfrak{a}$. Then $G^a G'^a = \sum_j \lambda_j^a F_j^{ha} = \sum_j \lambda_j^a F_j \in \mathfrak{a}$. Therefore say $G^a \in \mathfrak{a} \Longrightarrow G^{ah} \in \mathfrak{a}^h \Longrightarrow G = X_i^k G^{ah} \in \mathfrak{a}^h$.

h) Suppose $G^n \in \mathfrak{a}^h$, then $(G^a)^n \in \mathfrak{a}^{ha} = \mathfrak{a}$ whence $G^a \in \mathfrak{a}$ and $G^{ah} \in \mathfrak{a}^h \Longrightarrow G \in \mathfrak{a}^h$

■

Proposition 6.4.13 (Projective Space has Affine Covering)
Consider subsets $D_+(X_i)(K) = U_i(K) \subset \mathbb{P}_k^n(K)$ of the form

$$D_+(X_i)(K) = U_i(K) \coloneqq \{(x_0 : ... : x_n) \mid x_i \neq 0\} \quad i = 0, ..., n$$

These are open in the Zariski topology. Furthermore under the subspace topology U_i is homeomorphic to $\mathbb{A}_k^n(K)$, with an explicit homeomorphism given by

$$\theta_i : \mathbb{A}_k^n(K) \to U_i$$
$$(x_0, ..., x_{i-1}, x_{i+1}, ..., x_n) \to [x_0 : ... : 1 : ... : x_n]$$

For $F \in k[x_0, ..., x_{i-1}, x_{i+1}, ..., x_n]$ and $(x) \in U_i$ we have

$$F(x) = 0 \Longleftrightarrow F^h(\theta_i(x)) = 0$$

and similarly for $G \in k[X_0, ..., X_n]$ we have

$$G(\theta_i(x)) = 0 \Longleftrightarrow G^a(x) = 0$$

Further we have

a) $\theta_i\left(V^{\text{aff}}(\mathfrak{a})\right) = V_+\left(\mathfrak{a}^h\right) \cap U_i$ for $\mathfrak{a} \lhd k[x_0, ..., x_{i-1}, x_{i+1}, ..., x_n]$
b) $\theta_i\left(V^{\text{aff}}(\mathfrak{b}^a)\right) = V_+(\mathfrak{b}) \cap U_i = V_+\left(\mathfrak{b}^{ah}\right) \cap U_i$ for $\mathfrak{b} \lhd k[X_0, ..., X_n]$ homogenous
c) $I^{\text{aff}}(Y)^h = I_+(\theta_i(Y))$ for $Y \subset U_i$

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Proof. This follows largely from (6.4.12). The relations show that $\theta_i(D(F)) = D_+\left(F^h\right)$ and $\theta_i^{-1}(D_+(G)) = D(G^a)$, so that θ_i is a homeomorphism as required.

a) $x \in \theta_i\left(V^{\text{aff}}(\mathfrak{a})\right) \implies x \in U_i$ and $F(\theta_i^{-1}(x)) = 0 \quad \forall F \in \mathfrak{a} \implies x \in U_i$ and $F^h(x) = 0 \quad \forall F \in \mathfrak{a} \implies G(x) = 0 \quad \forall G \in \mathfrak{a}^h \implies x \in V_+\left(\mathfrak{a}^h\right) \cap U_i$. Conversely $x \in V_+\left(\mathfrak{a}^h\right) \cap U_i \implies G(x) = 0 \quad \forall G \in \mathfrak{a}^h \implies F(\theta_i^{-1}(x)) = 0 \quad \forall F \in \mathfrak{a}^{ha} = \mathfrak{a} \implies x \in \theta_i\left(V^{\text{aff}}(\mathfrak{a})\right)$

b) $\theta_i\left(V^{\text{aff}}(\mathfrak{b}^a)\right) = V_+\left(\mathfrak{b}^{ah}\right) \cap U_i$. It remains to show that $V_+(\mathfrak{b}) \cap U_i = V_+\left(\mathfrak{b}^{ah}\right) \cap U_i$. As (6.4.12) $\mathfrak{b} \subset \mathfrak{b}^{ah}$ then $V_+\left(\mathfrak{b}^{ah}\right) \cap U_i \subset V_+(\mathfrak{b}) \cap U_i$. Conversely if $x \in V_+(\mathfrak{b}) \cap U_i$ then $G(x) = 0 \implies x_i^k G^{ah}(x) = 0 \implies G^{ah}(x) = 0$ because $x_i \neq 0$. This shows $x \in V_+\left(\mathfrak{b}^{ah}\right) \cap U_i$

c) Suppose $G \in I^{\text{aff}}(Y)^h$ then $G = \sum_j \lambda_j F_j^h$ for $F_j \in I^{\text{aff}}(Y)$ and $\lambda_j \in k[X_0, ..., X_n]$. Then $F_j(x) = 0 \implies F_j^h(\theta_i(x)) = 0 \implies G(\theta_i(x)) = 0$. Therefore we conclude $G \in I_+(\theta_i(Y))$. Conversely if $G \in I_+(\theta_i(Y))$ then $G^a(x) = 0$ for all $x \in Y$ and $G^a \in I^{\text{aff}}(Y) \implies G^{ah} \in I^{\text{aff}}(Y)^h \implies G = X_i^k G^{ah} \in I^{\text{aff}}(Y)^h$.

■

Lemma 6.4.14
Let $X = V_+(\mathfrak{b})$ be a projective variety. Then $X \cap U_i \neq \emptyset \iff X_i \notin \mathfrak{b} \iff \overline{X_i} \neq 0$.

Proof. Observe $X \cap U_i = \emptyset \iff X \subset V_+(X_i) \iff X_i \in I_+(X) = \mathfrak{b}$. ■

Corollary 6.4.15
Let $X = V_+(\mathfrak{b}) \subset \mathbb{P}_k^n$ be a projective variety and for each affine chart $U_i \subset \mathbb{P}_k^n$ consider the affine variety $X^a := V^{\text{aff}}(\mathfrak{b}^a) \subset \mathbb{A}_k^n$. Then there is a natural homeomorphism

$$\theta_i : X^a(K) \xrightarrow{\sim} D_+\left(\overline{X_i}\right) = X(K) \cap U_i(K)$$

for all algebraic extensions K/k.

Proposition 6.4.16 (Projective Closure)
Identify $\mathbb{A}_k^n\left(\overline{k}\right)$ with $U_i \subset \mathbb{P}_k^n\left(\overline{k}\right)$. Let $Y \subset U_i$ be a subset. Then we have the following expression for topological closure

$$\overline{Y} = V_+\left(I^{\text{aff}}(Y)^h\right)$$
$$\overline{Y} \cap U_i = \text{cl}_{U_i}(Y)$$

*We call $\overline{Y}$ the **projective closure** of Y.*

Proof. By (6.4.13).c) $I_+(Y) = I^{\text{aff}}(Y)^h \implies \overline{Y} \overset{(6.4.4)}{=} V_+ I_+(Y) = V_+\left(I^{\text{aff}}(Y)^h\right)$.

Then $\text{cl}_{U_i}(Y) = \overline{Y} \cap U_i$ by (4.1.21) (or more directly $\overset{(6.1.2)}{=} V_+(I_+(Y)) \overset{(6.4.13).a)}{=} V_+\left(I^{\text{aff}}(Y)^h\right) \cap U_i = \overline{Y} \cap U_i$). ■

6.4.3 Galois Orbits

Proposition 6.4.17
Let $X = V_+(\mathfrak{a}) \subset \mathbb{P}_k^n$ be a projective variety and K/k a normal algebraic extension (e.g. $\overline{k}$). Then there is a well-defined group action

$$\text{Aut}(K/k) \times X(K) \to X(K)$$
$$(\sigma, (x_0 : ... : x_n)) \to (\sigma(x_0) : ... : \sigma(x_n))$$

which commutes with the affine charts

$$\sigma(\theta_i(x)) = \theta_i(\sigma(x)) \quad \forall x \in U_i(K) \cap X(K)$$

Write $X_0(K)$ for the set of equivalence classes under this action. The projection map

$$\pi : X(K) \to X_0(K)$$

is the Kolmogorov Quotient (4.1.36). In particular points $x, y \in X(K)$ are topologically indistinguishable iff they are Galois conjugate. Further closed (resp. open) subsets are Galois-equivariant.

Proof. First consider the case $X = \mathbb{P}_k^n$. If $(x_0 : ... : x_n) = (x_0' : ... : x_n')$ then $x_i = \lambda_i x_i'$ for $\lambda_i \in K^\star$. Then $\sigma(x_i) = \sigma(\lambda_i)\sigma(x_i')$ so $(\sigma(x_0) : ... : \sigma(x_n)) \sim (\sigma(x_0') : ... : \sigma(x_n'))$ and the action is well-defined. For $x \in X(K)$ write $\sigma(x)$. Then $\tau(\sigma(x)) = (\tau \circ \sigma)(x)$ so it is a group action.

For the general case if $F(x) = 0$ for $F \in k[X_0, ..., X_n]$ then $F(\sigma(x)) = 0$. Therefore $x \in V(\mathfrak{a})(K) \implies \sigma(x) \in V(\mathfrak{a})(K)$ and the action is well-defined on $X(K)$.

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We need to show that two points $x, y \in X(K)$ are conjugate by $\mathrm{Aut}(K/k)$ iff they are topologically indistinguishable. Suppose $x = \sigma(y)$ and $x \in V(\mathfrak{b})$. Then evidently $y \in V(\mathfrak{b})$. Conversely if they are topologically indistinguishable then we may suppose $x, y \in U_i$. Then $\theta_i(x), \theta_i(y)$ are also indistinguishable, so by (6.1.9) they are Galois conjugate, and so $\theta_i(x) = \theta_i(\sigma(y))$. As θ_i is bijective we deduce $x = \sigma(y)$ as required. ∎

Proposition 6.4.18
Let $X = V_+(\mathfrak{a}) \subset \mathbb{P}^n_k$ be a projective variety and $L/K/k$ a tower of normal extensions. Then there is an injective map

$$\begin{array}{ccc} X(K) & \xhookrightarrow{i_{KL}} & X(L) \\ \downarrow{\scriptstyle \pi} & & \downarrow{\scriptstyle \pi} \\ X_0(K) & \overset{i_0}{\dashrightarrow} & X_0(L) \end{array}$$

Proof. In order for i_0 to be well-defined, we require that $\pi(x) = \pi(y) \implies \pi(i_{KL}(x)) = \pi(i_{KL}(y))$. Recall $\pi(x) = \pi(y) \implies x = \sigma(y)$ for some $\sigma \in \mathrm{Aut}(K/k)$. Then $i_{KL} \circ \sigma \in \mathrm{Mor}_k(K, L)$ and by (3.17.71) there exists $\hat{\sigma} \in \mathrm{Aut}(L/k)$ such that $\hat{\sigma}|_K := \hat{\sigma} \circ i_{KL} = i_{KL} \circ \sigma$. Therefore $\hat{\sigma}(i_{KL}(x)) = i_{KL}(y) \implies \pi(i_{KL}(x)) = \pi(i_{KL}(y))$.

To show i_0 is injective we need to show the converse, which follows from (3.17.73). ∎

6.4.4 Projective Varieties are Abstract Varieties

Definition 6.4.19 (Regular Function)
Let $X = V_+(\mathfrak{a}) \subset \mathbb{P}^n_k$ be a projective variety and $U \subset X\left(\overline{k}\right)$ an open subset. We say a function

$$f : U \to \overline{k}$$

*is **regular** at $x \in U$ if there is a neighbourhood V of x such that*

$$f(y) = \frac{g(y)}{h(y)} \quad \forall\, y \in V$$

*for $g, h \in k[X]$ homogenous polynomials of the same degree with h nowhere zero on V. We therefore define the **structure sheaf** as follows*

$$\mathcal{O}_X(U) := \left\{ f : U \to \overline{k} \;\middle|\; f \text{ regular at all } x \in U \right\}$$

The pair $(X, \mathcal{O}_X)$ is a space of functions.

Proposition 6.4.20 (Structure Sheaf of Regular Functions)
Let $X = V_+(\mathfrak{a}) \subset \mathbb{P}^n_k$ be a projective variety, $U \subset X\left(\overline{k}\right)$ open and $f : U \to \overline{k}$ a function. Consider the affine charts

$$\theta_i : X_i^a\left(\overline{k}\right) \xrightarrow{\sim} D_+\left(\overline{X_i}\right) = X\left(\overline{k}\right) \cap U_i \quad i = 0, ..., n$$

where $X_i^a = V^{\mathrm{aff}}(\mathfrak{a}^a)$ as in (6.4.15). Then the following are equivalent

a) f is regular (at x) in the sense of (6.4.19)
b) $f \circ \theta_i : \theta_i^{-1}(U \cap U_i) \to \overline{k}$ is regular (at $\theta_i^{-1}(x)$) in the sense of (6.1.25) for all i with $U \cap U_i \neq \emptyset$ (for some, equivalently all, i with $x \in U \cap U_i$)

In other words for each $i = 0, ..., n$ there is an isomorphism of spaces with functions

$$\theta_i : \left(X_i^a\left(\overline{k}\right), \mathcal{O}_{X_i^a}\right) \xrightarrow{\sim} \left(D_+\left(\overline{X_i}\right), \mathcal{O}_X\,|_{D_+(\overline{X_i})}\right)$$

In particular $(X, \mathcal{O}_X)$ is an abstract variety.

Proof. By (6.4.15) each θ_i is a homeomorphism onto the open set $D_+\left(\overline{X_i}\right)$, and these cover $X\left(\overline{k}\right)$ as i varies.

$a) \implies b)$ Let $x := \theta_i(\hat{x})$. There exists an open nbhd V of x such that $f(y) = \frac{g(y)}{h(y)}$ for all $y \in V$. By definition $g = G + \mathfrak{a}$ and $h = H + \mathfrak{a}$ for $G, H \in k[X_0, ..., X_n]$ homogenous polynomials of the same degree, for which H doesn't vanish on V. Then $\hat{V} := \theta_i^{-1}(V)$ is a open nbhd of $\hat{x}$. For all $\hat{y} \in \hat{V}$ we have

$$(f \circ \theta_i)(\hat{y}) = f(\theta_i(\hat{y})) = \frac{G(\hat{y})}{H(\hat{y})} = \frac{G^a(\hat{y})}{H^a(\hat{y})}$$

with G^a, H^a polynomials in the affine coordinates, and H^a nonvanishing on $\hat{V}$.

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$b) \implies a)$ Suppose $x \in U \cap U_i$. We have by definition an open nbhd $\hat{V}$ of $\theta_i^{-1}(x)$ and $G, H \in k[x_0, x_1, ..., x_{i-1}, x_{i+1}, ..., x_n]$ for which

$$(f \circ \theta_i)(\hat{y}) = \frac{G(\hat{y})}{H(\hat{y})} \quad \forall \hat{y} \in \hat{V}$$

As θ_i is a homeomorphism, $V := \theta_i\left(\hat{V}\right)$ is an open nbhd of x. By (6.4.13) we have for $y := \theta_i(\hat{y})$

$$f(y) = \frac{G^h(y)}{H^h(y)}$$

which holds for all $y \in V$. Here we may multiply G^h or H^h by a power of X_i so that both have the same degree; this does not change the quotient on $D_+\left(\overline{X_i}\right)$ (where $X_i \neq 0$), and afterwards H^h does not vanish on V. ∎

We may generalise the dehomogenisation isomorphism of the previous result to an isomorphism of coordinate rings of the affine charts of an irreducible projective variety.

Proposition 6.4.21 (Structure Sheaf on Basic Affines)
Let $X = V_+(\mathfrak{p})$ be an irreducible projective variety and suppose $X \cap U_i \neq \emptyset$. Then there is an isomorphism of k-algebras

$$\begin{array}{ccc} \mathcal{O}_X\left(D_+\left(\overline{X_i}\right)\right) & \xrightarrow[\sim]{(6.4.20)} & \mathcal{O}_{X\cap U_i}(X \cap U_i) \\ \uparrow & & \uparrow{\scriptstyle (6.1.26)} \\ k[X]_{(\overline{X_i})} & \xrightarrow{\sim} & k[X \cap U_i] \end{array}$$

$$\frac{\overline{X_j}}{\overline{X_i}} \longleftrightarrow \overline{x_j}$$

and the dashed arrow is defined in the obvious way (and is therefore an isomorphism).

Proof. By (6.4.14) $X_i \notin \mathfrak{p}$. We first demonstrate the bottom map is well-defined and an isomorphism. Consider the following commutative diagram

$$\begin{array}{ccccc} & & k[X_0, ..., X_n]_{(X_i)} & \xrightarrow[\sim]{\theta_i^\sharp} & k[U_i] \\ & \swarrow & \downarrow{\scriptstyle \pi_1} & & \downarrow{\scriptstyle \pi_2} \\ k[X_0, ..., X_n]_{(X_i)}/\mathfrak{p}_{(X_i)} & \xrightarrow[(3.11.17)]{\sim} & k[X]_{(\overline{X_i})} & \overset{\sim}{\dashrightarrow} & k[X \cap U_i] \end{array}$$

where the isomorphism $\theta_i^\sharp$ is defined in (6.4.11).

It follows from (3.3.56) that π_1 is uniquely defined and has kernel $\mathfrak{p}_{(X_i)}$ which equals $\mathfrak{p}_{X_i} \cap k[X_0, ..., X_n]_{(X_i)}$ by (3.11.15). Therefore we may regard π_1 as a quotient map (it is surjective as every element of $k[X]_{(\overline{X_i})}$ is of the form $\overline{F}/\overline{X_i}^d$ with F homogenous of degree d).

We also claim that the diagonal map $\pi_2 \circ \theta_i^\sharp$ has kernel $\mathfrak{p}_{(X_i)}$. It is enough by (3.5.5).b) and the preceding comment to show that the composite map $\pi_2 \circ (-)^a : k[X_0, ..., X_n] \to k[X \cap U_i]$ has kernel $\mathfrak{p}$. However this follows from (6.4.12) for $F \in \ker(\pi_2 \circ (-)^a) \implies F^a \in \mathfrak{p}^a \implies F^a = G^a \implies F^{ah} = G^{ah} \in \mathfrak{p}^{ah} = \mathfrak{p} \implies F^{ah} \in \mathfrak{p} \implies X_i^k F \in \mathfrak{p} \implies F \in \mathfrak{p}$.

The required map exists by (3.3.56), as π_1 is a quotient map. Further it is injective, surjective and therefore an isomorphism (surjective as $\pi_2 \circ \theta_i^\sharp$ is, and injective as the kernels of π_1 and $\pi_2 \circ \theta_i^\sharp$ agree).

The map $k[X]_{(\overline{X_i})} \to \mathcal{O}_X\left(D_+\left(\overline{X_i}\right)\right)$ is $\overline{F}/\overline{X_i}^d \mapsto \left(y \mapsto F(y)/X_i(y)^d\right)$, which is well-defined and regular by definition. It is clear the diagram commutes and so this map is an isomorphism. ∎

Proposition 6.4.22
Let X be a projective variety. Then it is separated.

Proof. Drafted by Claude Sonnet, reviewed by David Rufino.

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By (6.4.20) the charts θ_i identify $X \cap U_i$ with the closed subvarieties X_i^a of the affine charts $U_i \cong \mathbb{A}_k^n$ of $\mathbb{P}_k^n$, so X is a closed subvariety of $\mathbb{P}_k^n$. Now $\mathbb{P}_k^n$ is separated by (6.2.97), and therefore so is X by (6.2.68). ∎

6.4.5 Regular Morphisms

For the category of quasi-projective varieties we introduce a number of equivalent definitions of morphism. In light of (6.4.20) then *b)* shows a function is regular iff it is regular in all affine coordinates.

Proposition 6.4.23 (Regular Morphism)
Let X be a projective variety, $U \subset X\left(\overline{k}\right)$ an open subset, Y a quasi-projective variety and $\phi : U \to Y\left(\overline{k}\right)$ a function. Then the following are equivalent

a) ***pulls back regular functions*** *ϕ is continuous and determines a morphism of sheaves*

$$\begin{aligned} \mathcal{O}_Y &\to \phi_\star \mathcal{O}_U \\ f &\to f \circ \phi \end{aligned}$$

b) ***affine projections are regular*** *There is an open cover $U = \bigcup_{i \in I} U_i$ such that $\phi(U_i)$ is a subset of an affine chart $W_{j(i)}$ and*

$$\pi_k \circ \theta_{j(i)}^{-1} \circ \phi|_{U_i} : U_i \to \overline{k}$$

is regular (6.4.19) for all $i \in I$ and $k = 1, ..., m$

c) ***locally homogenous*** *There is an open cover $U = \bigcup_{i \in I} U_i$ and families of homogenous polynomials $\{f_{i0}, ..., f_{im}\} \subset k[X]_{\text{hom}}$ of the same degree such that $U_i \subset D_+(f_{i0}) \cup ... \cup D_+(f_{im})$ and*

$$\phi(y) = (f_{i0}(y) : ... : f_{im}(y)) \quad \forall y \in U_i$$

In this case we say ϕ is ***regular*** *or a* ***regular morphism*** *of quasi-projective varieties.*

Proof. *b)* $\implies$ *a)* We may assume wlog that U_i is contained in an affine chart V_i. Therefore by definition and (6.4.20)

$$\theta_{j(i)}^{-1} \circ \phi \circ \theta_i \mid_{\theta_i^{-1}(U_i)} : \theta_i^{-1}(U_i) \to \mathbb{A}_k^m$$

is regular as a map of quasi-affine varieties. Therefore by (6.1.39) it is continuous. Whence $\phi|_{U_i}$ is continuous by (6.4.13), and ϕ is continuous by (4.1.27).

Suppose $f \in \mathcal{O}_Y(W)$ then it is enough to show for all $x \in \phi^{-1}(W)$ that $f \circ \phi$ is regular at x. Suppose $x \in U_i$ and replace W with $W \cap W_{j(i)}$. Then by definition $f \circ \theta_{j(i)} : \theta_{j(i)}^{-1}(W) \to \overline{k}$ is regular at $\theta_{j(i)}^{-1}(x)$. Whence by (6.1.40)

$$f \circ \phi \circ \theta_i = \left(f \circ \theta_{j(i)}\right) \circ \left(\theta_{j(i)}^{-1} \circ \phi \circ \theta_i\right)$$

is regular at $\theta_i^{-1}(x)$, and therefore by (6.4.20) $f \circ \phi$ is regular at x.

a) $\implies$ *b)* For $x \in U$ there is some affine W_j such that $\phi(x) \in W_j \implies x \in \phi^{-1}\left(W_j\right) =: V_x$. By (6.4.20) $\pi_k \circ \theta_j^{-1} : W_j \to \overline{k}$ is regular whence by assumption $\pi_k \circ \theta_j^{-1} \circ \phi|_{V_x}$ is regular. As every point has a nbhd V_x we may find the required open cover.

b) $\implies$ *c)* By definition there exists homogenous $f_{i0}, ..., f_{im}, g_{i0}, ..., g_{im} \in k[X]_{\text{hom}}$ such that

$$\phi(y) = \left(\frac{f_{i0}(y)}{g_{i0}(y)} : ... : 1 : ... : \frac{f_{im}(y)}{g_{im}(y)}\right) \quad \forall y \in U_i$$

Therefore we may define

$$h_{il} \coloneqq f_{il} \prod_{k \neq l} g_{ik}$$

Then evidently

$$\phi(y) = (h_{i0}(y) : ... : h_{im}(y)) \quad \forall y \in U_i$$

c) $\implies$ *b)* For every $x \in U$ there exists some open nbhd V_x such that

$$\phi(y) = (f_0(y) : ... : f_m(y)) \quad \forall y \in V_x$$

Suppose that $f_j(x) \neq 0$ then replace V_x with $V_x \cap D_+\left(f_j\right)$. Then

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$$\left(\pi_k \circ \theta_j^{-1} \circ \phi|_{V_x}\right)(y) = \frac{f_k(y)}{f_j(y)}$$

which is regular by definition. The family $\{V_x\}_{x\in U}$ is the corresponding open cover. ∎

Corollary 6.4.24
Let $\phi : X \to Y$ and $\psi : Y \to Z$ be regular morphisms of quasi-projective varieties. Then so is $\psi \circ \phi : X \to Z$.

Proof. This is immediate by (6.4.23).a). ∎

Corollary 6.4.25
A function $f : X\left(\overline{k}\right) \to \overline{k}$ is regular iff it coincides with a regular morphism $f : X\left(\overline{k}\right) \to D_+(X_0) \cong \mathbb{A}^1_k$. In particular a regular function is continuous with respect to the co-finite topology on $\overline{k}$.

Proof. This follows from (6.4.23).b). ∎

Proposition 6.4.26
Let $X \subset \mathbb{P}^n_k$ be a projective variety, $U \subset X\left(\overline{k}\right)$ an open subset and $U = \bigcup_{i\in I} U_i$ an open cover. Suppose that we have $f_{i0}, ..., f_{im} \in k[X]_{\text{hom}}$ homogenous of the same degree such that

a) For every $i \in I$

$$U_i \subset D_+(f_{i0}) \cup ... \cup D_+(f_{im})$$

b) For every pair $U_i \cap U_{i'} \neq \emptyset$ we have

$$f_{ik'}(y)f_{i'k}(y) = f_{ik}(y)f_{i'k'}(y) \quad \forall y \in U_i \cap U_{i'}, \forall k, k' = 0, ..., m$$

Then there exists a unique regular morphism $f : U \to \mathbb{P}^m_k\left(\overline{k}\right)$ such that

$$f(x) = (f_{i0}(x) : ... : f_{im}(x)) \quad \forall x \in U_i$$

6.4.6 Local Ring

Proposition 6.4.27
Let $X = V_+(\mathfrak{b}) \subset \mathbb{P}^n_k$ be a projective variety and $W = V_+(\mathfrak{p})$ an irreducible subset. Then there is an isomorphism

$$\begin{aligned} k[X]_{(\mathfrak{p}/\mathfrak{b})} &\xrightarrow{\sim} \mathcal{O}_{X,W} \\ \frac{\overline{F}}{\overline{G}} &\to \left[\left(D\left(\overline{G}\right), \frac{F(-)}{G(-)}\right)\right] \end{aligned}$$

More precisely for any affine chart U_i such that $U_i \cap W \neq \emptyset$ there is a commutative diagram

$$\begin{array}{ccccc} k[U_i]_{\mathfrak{p}^a} & \xrightarrow[(3.5.45)]{\sim} & k[U_i \cap X]_{\mathfrak{p}^a/\mathfrak{b}^a} & \xrightarrow[(6.1.56)]{\sim} & \mathcal{O}_{U_i\cap X, U_i\cap W} \\ \wr\downarrow & & & & \downarrow\wr \ (4.3.19) \\ k[X_0, ..., X_n]_{(\mathfrak{p})} & \xrightarrow[(3.11.18)]{\sim} & k[X]_{(\mathfrak{p}/\mathfrak{b})} & \dashrightarrow^{\sim} & \mathcal{O}_{X,W} \end{array}$$

Proof. We construct the left vertical map, and then the dashed arrow is uniquely defined.

As $U_i \cap W \neq \emptyset$ then $W \not\subset V(X_i) \implies X_i \notin \mathfrak{p}$. So by (6.4.12).d) $\mathfrak{p}^{ah} = \mathfrak{p}$. To complete the diagram we consider the mutually inverse maps

$$\begin{aligned} k[U_i] = k[x_0, ..., x_{i-1}, x_{i+1}, ..., x_n]_{\mathfrak{p}^a} &\to k[X_0, ..., X_n]_{(\mathfrak{p})} \\ \frac{F}{F'} &\to \frac{F\left(\frac{X_0}{X_i}, ..., \frac{X_n}{X_i}\right)}{F'\left(\frac{X_0}{X_i}, ..., \frac{X_n}{X_i}\right)} \\ \frac{G^a}{G'^a} &\leftarrow \frac{G}{G'} \end{aligned}$$

Note $F'^h \in \mathfrak{p} \implies F' = F'^{ha} \in \mathfrak{p}^a$. Therefore $F' \notin \mathfrak{p}^a \implies F'^h = X_i^{\deg(F)} F'\left(\frac{X_0}{X_1}, ..., \frac{X_n}{X_i}\right) \notin \mathfrak{p} \implies F'\left(\frac{X_0}{X_1}, ..., \frac{X_n}{X_i}\right) \notin \mathfrak{p}$. Similarly $G'^a \in \mathfrak{p}^a \implies G'^{ah} \in \mathfrak{p}^{ah} = \mathfrak{p} \implies G' = X_i^k G'^{ah} \in \mathfrak{p}$, so $G' \notin \mathfrak{p} \implies G'^a \notin \mathfrak{p}^a$. Finally

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$$\frac{G^a\left(\frac{X_0}{X_i},...,\frac{X_n}{X_i}\right)}{G'^a\left(\frac{X_0}{X_i},...,\frac{X_n}{X_i}\right)} = \frac{X_i^{-\deg(G)}G}{X_i^{-\deg(G')}G'} = \frac{G}{G'}$$

so the maps are mutually inverse. ∎

Corollary 6.4.28
Let X be an irreducible projective variety with $W = V_+(\mathfrak{p})$ an irreducible subvariety. Then there is a commutative diagram

$$\begin{array}{ccc} k[U_i]_{\mathfrak{p}^a} & \hookrightarrow & k[U_i]_{(0)} \\ \downarrow\sim & & \downarrow\sim \\ k[X]_{(\mathfrak{p})} & \hookrightarrow & k[X]_{((0))} \\ \downarrow\sim & & \downarrow\sim \\ \mathcal{O}_{X,W} & \hookrightarrow & k(X) \end{array}$$

In particular $k(X)$ is the field of fractions for $\mathcal{O}_{X,W}$.

Proof. The vertical maps are from (6.4.27) and the horizontal maps are the obvious ones. As the top map is injective and corresponds to field of fractions we deduce that $k(X)$ is field of fractions for $\mathcal{O}_{X,W}$. ∎

6.4.7 Dimension

Proposition 6.4.29
Projective space $\mathbb{P}_k^n$ has dimension n. Furthermore it is non-singular, normal and regular.

Proof. Projective space is integral (6.4.8) and has an open subset isomorphic to $\mathbb{A}_k^n$. So we are done by (6.2.57) and (6.1.50).

The conditions are local (6.2.84) (3.22.10). ∎

Proposition 6.4.30
Let $X = V_+(\mathfrak{p})$ be an irreducible projective variety then $\dim k[X] = \dim X + 1$.

Proof. We have $\overline{X_i} \neq 0$ for some i as X is non-empty. Therefore

$$\dim k[X] \overset{(3.32.31)}{=} \dim k[X]_{\overline{X_i}} \overset{(3.12.8)}{=} \dim k[X]_{(\overline{X_i})}[T,T^{-1}] \overset{(3.9.3)}{=} \dim k[X]_{(\overline{X_i})}[T]_T \overset{(3.32.31)}{=} \dim k[X]_{(\overline{X_i})}[T] \overset{(3.32.29)}{=} \dim k[X]_{(\overline{X_i})} + 1$$

By (6.4.21) $\dim k[X]_{(\overline{X_i})} = \dim k[X \cap U_i] = \dim X \cap U_i$.

Finally this equals $\dim X$ (4.1.88). ∎

Proposition 6.4.31
Let X be a projective variety and $x \in X\left(\overline{k}\right)$. Then

$$\overline{\{x\}} = \left\{\sigma(x) \;\middle|\; \sigma \in \mathrm{Aut}\left(\overline{k}/k\right)\right\}$$

is an irreducible set of order $\left[k\left(\mathfrak{m}_{X,x}\right) : k\right]_s$. Further X is a symmetric space.

Proof. We have $x \in U_i$ for some affine chart then we may use (6.4.20) to reduce to the affine case (6.1.18). ∎

Corollary 6.4.32
Let X be a projective variety and $Z \subset X$ a closed subset. Then the following are equivalent

a) $\dim Z = 0$
b) *Z is finite*

More precisely

$$Z = \overline{\{x_1\}} \cup ... \cup \overline{\{x_n\}}$$

for a finite number of points $x_1, ..., x_n \in Z$.

Proof. This follows from (4.1.91) and (6.4.31). ∎

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Proposition 6.4.33
Let $X = \mathbb{P}^n_k\left(\overline{k}\right)$ with $n \geq 1$ and $G \in k[X_0, ..., X_n]$ an irreducible, homogenous polynomial of positive degree. Then $\dim V_+(G) = n - 1$.

Proof. Drafted by Claude Sonnet, reviewed by David Rufino.

Let $Y \coloneqq V_+(G)$. We first show $Y \neq \emptyset$. The affine cone $V(G) \subset \mathbb{A}^{n+1}_k$ is a hypersurface in an integral affine variety of dimension $n+1$, so every irreducible component has dimension n by (6.1.53), and $V(G)$ contains 0 as G is homogenous of positive degree. As $n \geq 1$ the set $V(G)$ is therefore not just $\{0\}$, and any non-zero point $(x_0, ..., x_n) \in V(G)$ gives a point $(x_0 : ... : x_n) \in Y$.

Now for each i we have $Y \cap U_i = V(G^a)$ (6.4.13), where $G^a \neq 0$ as $G = X_i^m G^{ah}$. By (6.1.53) applied to $\mathbb{A}^n_k$, every irreducible component of $Y \cap U_i$ has dimension $n-1$, so $\dim Y \cap U_i = n - 1$ whenever $Y \cap U_i \neq \emptyset$. Since the U_i form an open cover and $Y \neq \emptyset$, we conclude $\dim Y = n - 1$ by (4.1.88). ∎

6.4.8 Valuative Completeness

Lemma 6.4.34
Let R be a valuation ring for K. Suppose $x_0, ..., x_n \in K$ not all zero, then there exists $0 \leq j \leq n$ such that $\frac{x_i}{x_j} \in R$ for all $i = 0, ..., n$.

Proof. Recall (3.23.17) that the abelian group $K^\star/R^\star$ is totally ordered by the condition $x \leq y \iff yx^{-1} \in R^\star$. Choose j for which $x_j \neq 0$ such that x_j is minimal under this ordering. ∎

Proposition 6.4.35 (Projective Varieties are Complete)
Let $X = V_+(\mathfrak{p}) \subset \mathbb{P}^n_k$ be an irreducible projective variety. Let $k \subset R \subset k(X)$ be a valuation ring. Then there exists a point $x \in X\left(\overline{k}\right)$ such that $\mathcal{O}_{X,x} \subset R$ and $\mathfrak{m}_R \cap \mathcal{O}_{X,x} = \mathfrak{m}_{X,x}$.

More precisely there exists a local homomorphism $\Phi : R \to \overline{k}$ such that $\Phi|_{\mathcal{O}_{X,x}} = \mathrm{ev}_x$.

Proof. Denote the homogenous coordinate ring by simply $k[X]$. In light of (6.4.28) we may assume that $k(X) = k[X]_{((0))}$ and $\mathcal{O}_{X,x} = k[X]_{\mathfrak{p}_x}$ with $\mathfrak{m}_{X,x} = \left(\mathfrak{p}_x k[X]_{\mathfrak{p}_x}\right)_{(0)}$.

By (...) we may assume that $k(\mathfrak{m}_R)/k$ is algebraic and therefore by (3.17.63) there exists a local k-algebra homomorphisms

$$\Phi : R \to k(\mathfrak{m}_R) \hookrightarrow \overline{k}$$

We may assume wlog that $X \cap U_0 \neq \emptyset$ and therefore $X_0 \notin \mathfrak{p}$ where $X_0, ..., X_n$ are the homogenous coordinates. Define $x_i \coloneqq \frac{\overline{X}_i}{\overline{X}_0} \in k[X]_{((0))}$ for $i = 0, ..., n$. By (6.4.34) there is $1 \leq j \leq n$ such that $t_i \coloneqq \frac{x_i}{x_j} = \frac{\overline{X}_i}{\overline{X}_j} \in R$ for all $i = 0, ..., n$.

Note that $t_j = 1 \notin \mathfrak{m}_R$ so $\Phi\left(t_j\right) \neq 0$ we have a well-defined projective point $x = [\Phi(t_0) : ... : \Phi(t_n)] \in \mathbb{P}^n_k\left(\overline{k}\right)$. For any homogenous polynomial $F \in k[X_0, ..., X_n]$ of degree d we have

$$\begin{aligned} F(t_0, ..., t_n) &= \overline{X}_j^d F\left(\overline{X}_0, ..., \overline{X}_n\right) \\ &= \overline{X}_j^d \overline{F} \end{aligned}$$

regarding $k(X)$ as a subring of $\mathrm{Frac}(k[X])$ and $\bar{\cdot}$ denoting the reduction map $k[X_0, ..., X_n] \to k[X]$. Further

$$F(\Phi(t_0), ..., \Phi(t_n)) = \Phi(F(t_0, ..., t_n))$$

If $F \in \mathfrak{p}$ then $\overline{F} = 0$ and therefore $F(x) = 0$. This shows that $x \in X\left(\overline{k}\right)$.

We require to show that $\mathcal{O}_{X,x} \subset R$ and $\Phi|_{\mathcal{O}_{X,x}} = \mathrm{ev}_x$, from which it immediately follows that $\mathfrak{m}_R \cap \mathcal{O}_{X,x} = \mathfrak{m}_{X,x}$

Consider $g, h \in k[X]$ homogenous of the same degree with $h \notin \mathfrak{p}_x$. We claim that $g/h \in R$. For there exists $G, H \in k[X_0, ..., X_n]$ homogenous such that

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$$\begin{aligned} g &= G\left(\overline{X_0}, ..., \overline{X_n}\right) \\ &= \overline{X}_j^d G(t_0, ..., t_n) \\ h &= H\left(\overline{X_0}, ..., \overline{X_n}\right) \\ &= \overline{X}_j^d H(t_0, ..., t_n) \\ \frac{g}{h} &= \frac{G(t_0, ..., t_n)}{H(t_0, ..., t_n)} \end{aligned}$$

Evidently $G(t_0, ..., t_n) \in R$ since R is a k-algebra. Further by assumption

$$\begin{aligned} 0 \neq h(\Phi(t_0), ..., \Phi(t_n)) \\ = H(\Phi(t_0), ..., \Phi(t_n)) \\ = \Phi(H(t_0, ..., t_n)) \end{aligned}$$

which means $H(t_0, ..., t_n) \notin \mathfrak{m}_R \implies H(t_0, ..., t_n) \in R^\star$. Using the expression above for $\frac{g}{h}$ shows it lies in R as required. Further

$$\begin{aligned} \Phi\left(\frac{g}{h}\right) &= \frac{\Phi(G(t_0, ..., t_n))}{\Phi(H(t_0, ..., t_n))} \\ &= \frac{G(\Phi(t_0), ..., \Phi(t_n))}{H(\Phi(t_0), ..., \Phi(t_n))} \\ &= \frac{G(x)}{H(x)} \\ &= \frac{g}{h}(x) \end{aligned}$$

as required. ■

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6.5 Arithmetic Varieties

Remark 6.5.1
This section is rough working notes rather than a polished treatment, kept here for personal reference. Omitted from the HTML editions.

Proposition 6.5.2
Let $\mathfrak{o}$ be a ring with quotient field k. For an ideal $\mathfrak{a} \lhd \mathfrak{o}[X_1, ..., X_n]$ define the zero-locus

$$|X| := V(\mathfrak{a}) := \left\{ x \in \overline{k}^n \;\middle|\; f(x_1, ..., x_n) = 0 \quad \forall f \in \mathfrak{a} \right\}$$

and the coordinate ring $\mathfrak{o}[X] := \mathfrak{o}[X_1, ..., X_n]/\mathfrak{a}$. Then

$$|X| = V(\mathfrak{a}_k) \subset \overline{k}^n$$

$$\mathfrak{a}_k := \left\{ \sum_{i=1}^{n} \lambda_i f_i \in k[X_1, ..., X_n] \;\middle|\; f_i \in \mathfrak{a}, \lambda_i \in k \right\}$$

where $\mathfrak{a}_k \lhd k[X_1, ..., X_n]$. For every $f \in \mathfrak{o}[X_1, ..., X_n]/\mathfrak{a}$ the sets of the form

$$D(f) := \{x \in |X| \mid f(x) \neq 0\}$$

form a basis for the usual Zariski topology. For every principal open set $D(f) \subset |X|$ define

$$\mathcal{O}_X(D(f)) := \left\{ \phi : D(f) \to \overline{k} \;\middle|\; \phi(x) = \frac{g(x)}{f^r(x)} \forall x \in D(f), \text{some } g \in \mathfrak{o}[X], r \geq 0 \right\}$$

then $\mathcal{O}_X$ extends uniquely to sheaf of $\overline{k}$-valued functions with the obvious $\mathfrak{o}$-algebra structure.

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6.6 Scheme Notes

Remark 6.6.1
This section is rough working notes rather than a polished treatment, kept here for personal reference. Omitted from the HTML editions.

6.6.1 Schemes over $\mathbb{Z}$

Let A be a finitely-generated $\mathbb{Z}$-algebra, it may be expressed in the form $A = \mathbb{Z}[X_1, ..., X_n]/\mathfrak{a}$. For any $\mathbb{Q}$-algebra R the rational points are given by $\mathrm{AlgHom}_{\mathbb{Z}}(A, [R]_{\mathbb{Z}}) = \mathrm{AlgHom}_{\mathbb{Q}}(A \otimes_{\mathbb{Z}} \mathbb{Q}, R)$.

6.6.2 Closed Points vs Maximal Ideals

In the case of affine schemes there is a straightforward relationship between closed points and maximal ideals.

Proposition 6.6.2
Let $X = \mathrm{Spec}(A)$ be an affine scheme. Then the closed points of X correspond precisely to maximal ideals of A.

For a general scheme a closed point corresponds to a maximal ideal of all affine neighbourhoods.

Proposition 6.6.3
Let X be a scheme, $x \in X$ and $U \subset X$ an affine neighbourhood, with $U \xrightarrow{\sim} \mathrm{Spec}(A)$. Then if x is a closed in X we have x is closed in U and corresponds to a maximal ideal of A.

Remark 6.6.4
However the converse may not hold – there may be maximal ideals of affines subsets which do not correspond to a closed point. As an extreme example it's possible to construct a scheme with no closed points at all. Consider a local ring A with the following ideal structure

$$0 \subsetneq \mathfrak{p}_0 \subsetneq ... \subsetneq \mathfrak{p}_n \subsetneq ... \subsetneq \mathfrak{m} \subsetneq A$$

Then $\mathrm{Spec}(A) \setminus \{[\mathfrak{m}]\}$ is an open subscheme with no closed points.

More simply consider the spectrum of a discrete valuation ring $\mathrm{Spec}(A) = \{(0), \mathfrak{m}\}$. Then $\{(0)\}$ is not closed in $X := \mathrm{Spec}(A)$, but it is open (being the complement of $V(\mathfrak{m})$) and even isomorphic to $\mathrm{Spec}\big(A_{(0)}\big)$, and so (0) is closed in an affine neighbourhood.

However we may obtain some partial converses

Proposition 6.6.5
Let X be a quasi-compact scheme, $U \subset X$ an affine open neighbourhood and $x \in U$ a closed point. Then there exists $y \in \overline{\{x\}}$ such that y is closed.

Proposition 6.6.6
Let X be a integral scheme locally of finite type over k. Then if $x \in U$ is closed for U affine, it is also closed in X.

Proof. By assumption we may write $X = \bigcup_{i \in I} V_i$ where V_i are affine (as finite type $\implies$ quasi-compact). Then by (4.1.21)

$$\mathrm{cl}_X(\{x\}) \cap V_i = \mathrm{cl}_{V_i}(\{x\})$$

whenever $x \in V_i$ (and it is empty otherwise). Therefore it is sufficient to show that $x \in V_i \implies x$ closed in V_i.

So we may consider the following situation: $U = \mathrm{Spec}(A)$ and $V = \mathrm{Spec}(B)$ are affine open neighbourhoods of x, for A, B finitely generated k-algebras, where additionally x is closed in U. Then by (6.6.2) x corresponds to a maximal ideal $\mathfrak{m}$ of A, and a prime ideal of $\mathfrak{p}$ of B. As these are both stalks of X we have a local k-algebra isomorphism $A_{\mathfrak{m}} \xrightarrow{\sim} B_{\mathfrak{p}}$, and therefore also of residue fields $k(\mathfrak{m}) \xrightarrow{\sim} k(\mathfrak{p})$. Therefore by following lemma we have x is closed in V as required. ∎

Lemma 6.6.7
Let A be a finitely-generated k-algebra and $\mathfrak{p}$ a prime ideal. Then the following are equivalent

a) $\mathfrak{p}$ is maximal
b) $[\mathfrak{p}]$ is a closed point of $\mathrm{Spec}(A)$.
c) $k(\mathfrak{p})/k$ is algebraic
d) $k(\mathfrak{p})/k$ is finite

In particular the closed points of $\mathrm{Spec}(A)$ correspond precisely to the maximal ideals of A.

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Proof. First we observe that $k(\mathfrak{p})/k$ is always finitely generated, so $c) \iff d)$ is simply (3.17.44). Further $a) \iff b)$ is simply (6.6.2).

$d) \implies a)$ We have an inclusion $A/\mathfrak{p} \hookrightarrow k(\mathfrak{p})/k$. We deduce that $A/\mathfrak{p}$ is a finite-dimensional, integral k-algebra and therefore a field (3.17.47).

$a) \implies d)$ As A is finitely generated, so is $k(\mathfrak{p})/k$. By Zariski's Lemma (3.32.23) it is finite. ∎

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